

The VIA eRev truck is a factory-new, electrified, extended range vehicle. This course is designed to familiarize technicians with the truck's components and systems, as well as provide an overview of important truck safety information.

https://vimeo.com/136007732

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COURSE OVERVIEW

This course will focus on the following components:

- High Voltage safety overview
- Driver interface
- Fluid fill locations
- Power electronics cooling system
- Generator and Traction motor cooling system
- Charging system
- Export power panel
- Drive system
- Low Voltage fuse center Hybrid Control Unit (HCU)
- HVAC system
- Maintenance intervals

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HIGH VOLTAGE OVERVIEW

- High voltage is normally considered to be any voltage over 50 volts either AC or DC. The high voltage battery in a VIA extended range vehicle is approximately 400 volts, so all safety precautions must be taken when servicing the vehicle.
- When the EVSE is connected, or when export power is in use, up to 240V A/C and 400V D/C will be present on certain circuits.
- A VIA battery that shows a depleted state of charge, still carries at least a 20% charge and has the capacity to produce an electrical shock.
- Be aware of high voltage safety warning labels on components.
- High voltage circuits will be identified by orange cables or conduit, but all circuits should be treated as potentially dangerous. Never cut or open high voltage components and orange cables. Failure to follow this warning could result in property damage, personal injury, or death.
- Heavy gauge wire is not always required for high voltage flow. ALL orange wiring should be assumed to be carrying high voltage.
- A 12 volt signal sent to the Battery Control Module (or BCM) closes contactors allowing voltage to flow to the high voltage system.
- Always follow the “High Voltage Disabling/Enabling” procedure before servicing the vehicle.

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DRIVER INTERFACE

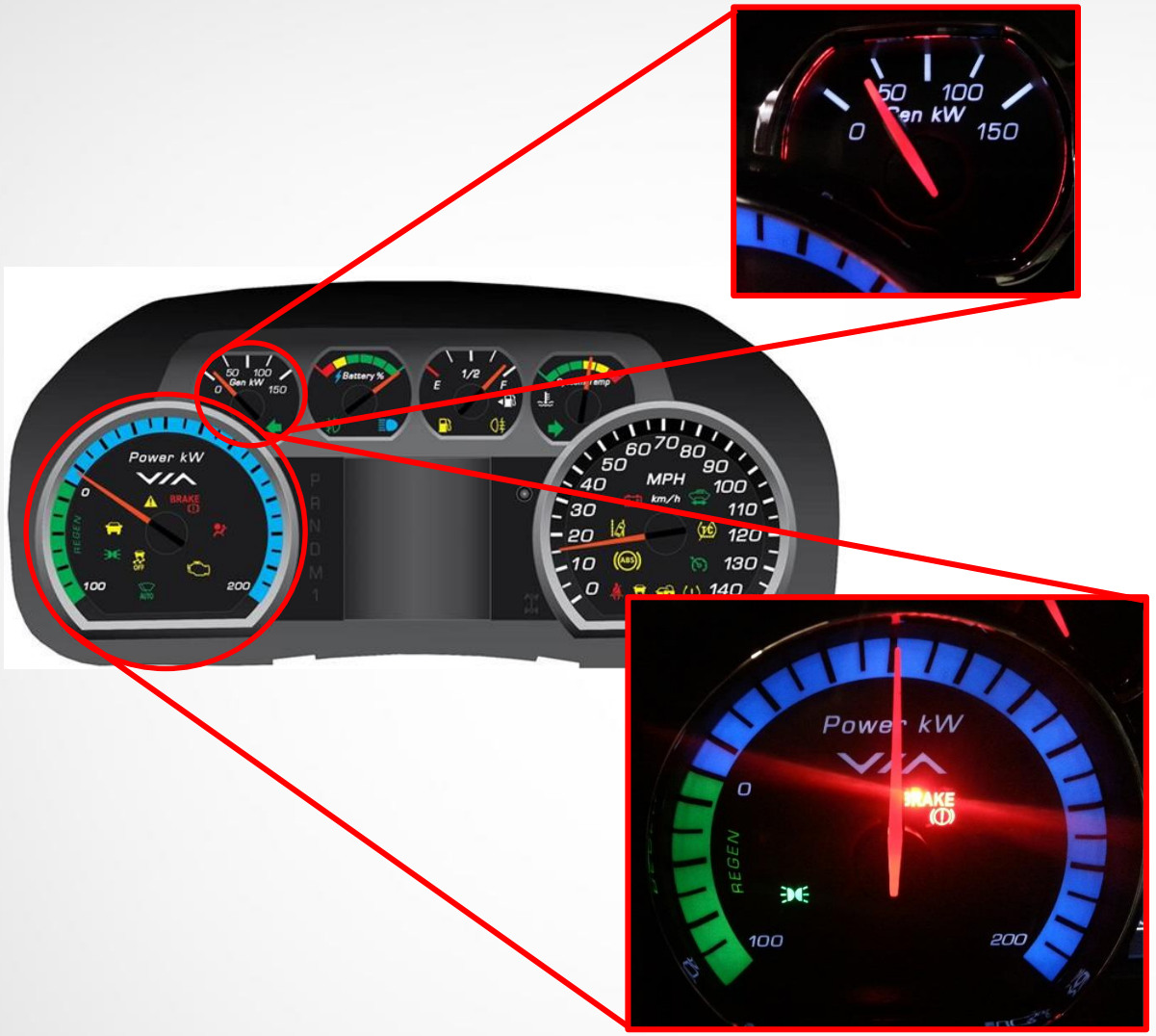


- The **Battery Percent Gauge** indicates the charge in the High Voltage battery.
- When the EVSE is connected and charging is complete, the gauge will indicate 100% to reflect a fully-charged high voltage battery. May also show 80% charge if vehicle is in purge mode.
- As the vehicle is driven in battery electric mode, the percentage will reduce until the engine starts. At that point the vehicle will be powered by the engine mounted generator.

- The **Drive Ready indicator** will illuminate green when the vehicle is ready to drive.
- It is used while in battery electric startup mode to indicate all necessary systems are functioning and the vehicle is ready to drive.
- The indicator will stay on as long as the vehicle is active.

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DRIVER INTERFACE



- The **Generator Kilowatt** gauge shows the output of the engine mounted generator.
- Shows high voltage output during export power mode.

- The **Power Kilowatt gauge** has a dual purpose.
- While accelerating or driving, the blue scale will show power usage.
- When coasting or braking, the green scale will show regenerative power recovery.

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DRIVER INTERFACE



- The **Hybrid System Lamp** (HSL) illuminates when the VIA system detects a malfunction.
- The HSL will begin to flash on and off if it detects a severe malfunction.
- Depending on the severity, the vehicle performance may be automatically reduced to protect itself.

- The **Malfunction Indicator Lamp** (or MIL) illuminates amber when the vehicle is placed in ON/RUN, as a check to confirm functionality
- If the MIL stays on for more than 15 seconds on startup or while the engine is running, this indicates that the GM OBD 2 system has detected a malfunction.

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SHIFT SELECTION



The gear selector works the same as it would in any conventional vehicle (P is park, R is reverse..etc.) because the vehicle does not have a transmission these actions are performed differently:

- Park is achieved with a module on the gearbox called a park pawl, which manually locks and unlocks the drivetrain.
- For Reverse, Neutral and Drive a signal is sent to the traction inverter to indicate direction of motor rotation.
- When the vehicle is placed into low gear (1 or M1) the APU is enabled and will generate power according to demand and maintain current battery state of charge. This will also provide additional power during hill climbs and cold battery operation.

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VAPOR CANISTER PURGE

If the vehicle has been used only in battery electric mode or the engine has not started for 5 days, the engine will automatically start when the vehicle is powered up to purge the vapor canister. The vehicle must be driven for approximately 15 to 30 minutes (generally at or above 55mph) to purge the canister. Once the canister is purged the vehicle may return to battery electric mode.

The battery will not charge above 80% if a canister purge is required, as the engine cannot run if the vehicle is above an 80% charge.

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VAPOR CANISTER PURGE

- *The onboard evaporative emission (EVAP) control system limits harmful fuel vapors from escaping into the atmosphere. During periods where the vehicle is not driven and battery-electric driving, fuel vapors accumulate and are stored in the EVAP canister until the engine is able to use them. After not using or driving the vehicle without sufficient engine operation for long enough, the EVAP canister is almost saturated with vapors and the vehicle control system must run the engine to purge and prevent overflow into the atmosphere.*

- *Purge operation will be requested after 4.7 days without sufficient engine operation and activated once the battery is below 80% charge. If the drive starts with the battery still above this level, the vehicle will operate in electric mode until the battery depletes to 80%, at which point purge operation will begin. Purge operation is similar to normal charge sustaining operation except for possible elevated battery charge level and a maximum power of 45 kW. In extreme circumstances, this can result in reduced performance while purge operation is active. If a refueling event occurs, the vehicle control system will reduce the 4.7-day period to account for additional vapors pushed into the canister. At most, if the tank is refilled completely from empty, purge operation will be requested as much as 15 hours before normal.*
- *EVAP canister purge is completed during a drive if the engine runs at sufficient load for at least 15 minutes. This 15 minutes may be achieved over any number of drives within 2 days. If 15 minutes of sufficient engine load is not achieved within the 2- day period, the timer will reset and 15 minutes will again be required. Sufficient engine load generally occurs when driving at speeds greater than 50 mph (i.e. highway driving).*

- *If the battery is charged during the period when EVAP canister purge is required (after 4.7 days without sufficient engine operation), the vehicle control system will not allow it to charge above 80%. This ensures the engine will immediately be able to run and consume all the vapors at the beginning of the drive.*

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CHECKING FLUIDS - FILL LOCATIONS



Power Electronics Coolant Reservoir

Generator, Traction Motor Coolant/Lubrication Reservoir

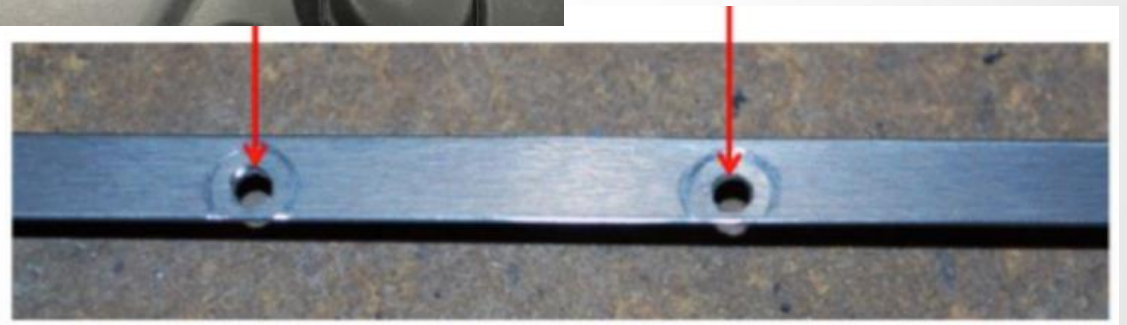
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CHECKING FLUIDS – GENERATOR / TRACTION MOTOR

- Flip the handle up, pull out the dipstick, and wipe it with a clean rag or paper towel.
- Push it back in all the way, wait three seconds, and pull it back out again.
- Check both sides of the dipstick, and read the lower level. The fluid level must be between the 2 indicators on the dipstick. Be sure to keep the dipstick pointed down to get an accurate reading.
- If the fluid level is in the acceptable range, push the dipstick back in all the way, then flip the handle down to lock the dipstick in place.

Note: The Generator and Traction Motor fluid should be checked weekly and when the fluid is cold. Use only DEX VI Automatic Transmission Fluid





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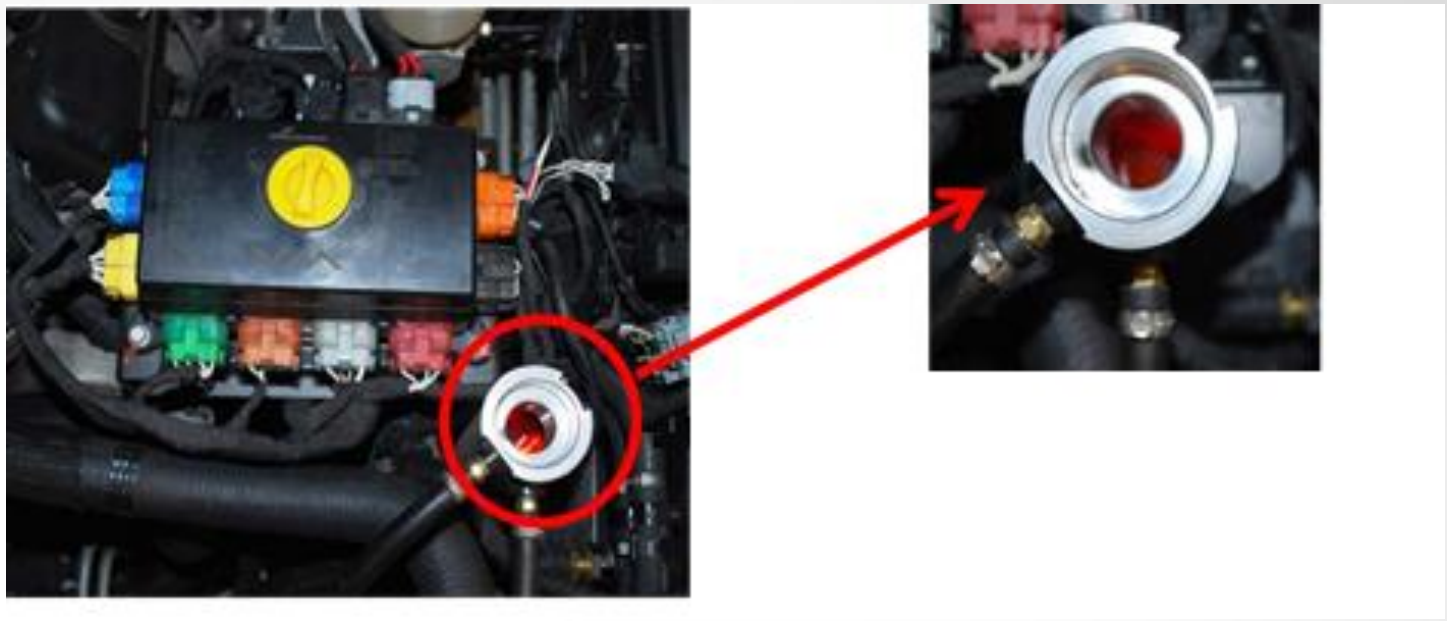
CHECKING FLUIDS – POWER ELECTRONICS

The power electronics fluid level should be checked weekly.

- Remove the surge tank pressure cap when the cooling system is no longer pressurized.
- Push down the pressure cap as you turn it.
- Remove the pressure cap.
- Fill the reservoir with the proper mixture of 50% DEX-COOL and deionized water, 2 inches below the top of the reservoir.
- Replace the pressure cap.

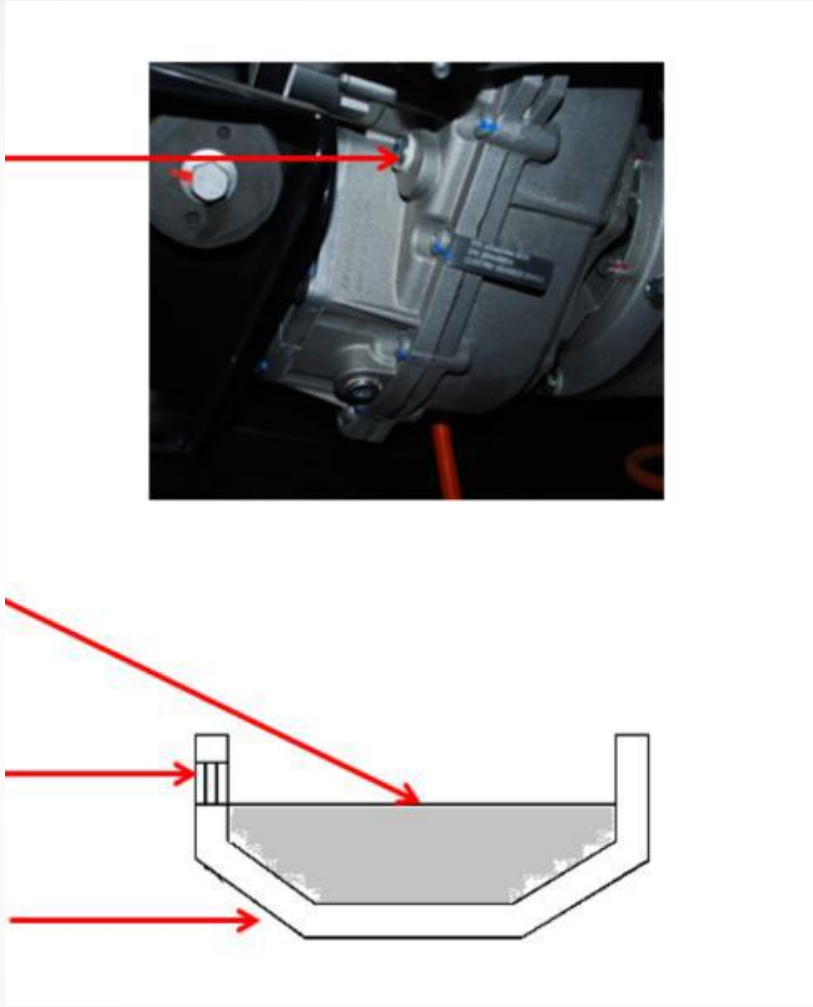
Warning: Steam and scalding liquids from a hot cooling system can blow out and cause serious burns. Never turn the surge tank pressure cap, when the cooling system is hot. Wait for the cooling system and reservoir tank pressure cap to cool.

Note: the GM coolant system also need to be checked on a regular basis.



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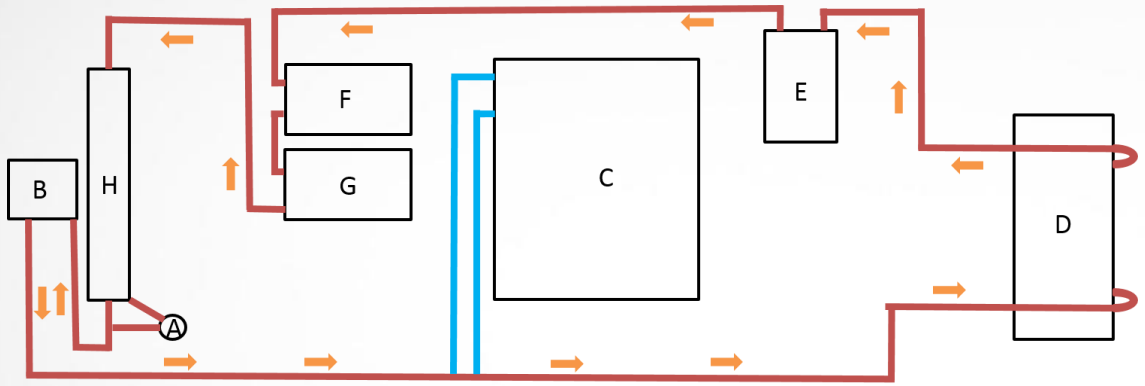
CHECKING FLUIDS - GEARBOX OIL LEVEL



- The gearbox fluid level should be checked at every engine oil change interval.
- The fluid level should be checked with the vehicle parked on a level surface.
- With the fill plug removed, the fluid level should be even with the bottom of the case fill plug opening.
- Use only DEX VI Automatic Transmission Fluid

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POWER ELECTRONICS COOLING SYSTEM



A. Reservoir
B. Coolant Pump
C. HV Battery
D. Bi-Directional Charger
E. DC/DC
F. Generator Inverter
G. Traction Inverter
H. Radiator

The power electronics cooling system uses a 50-50 mix of DEX-COOL and Deionized water to cool the following components, Generator and Traction inverters, High Voltage Battery, DC to DC Converter, and Bi-Directional Charger. With a capacity of approximately 2 U.S. gallons the system is comprised of a radiator, coolant pump, reservoir, and cooling fans. Note: The pump is located near the center of the vehicle above the front splash guard. It is normal for the pump to be heard running during charging.

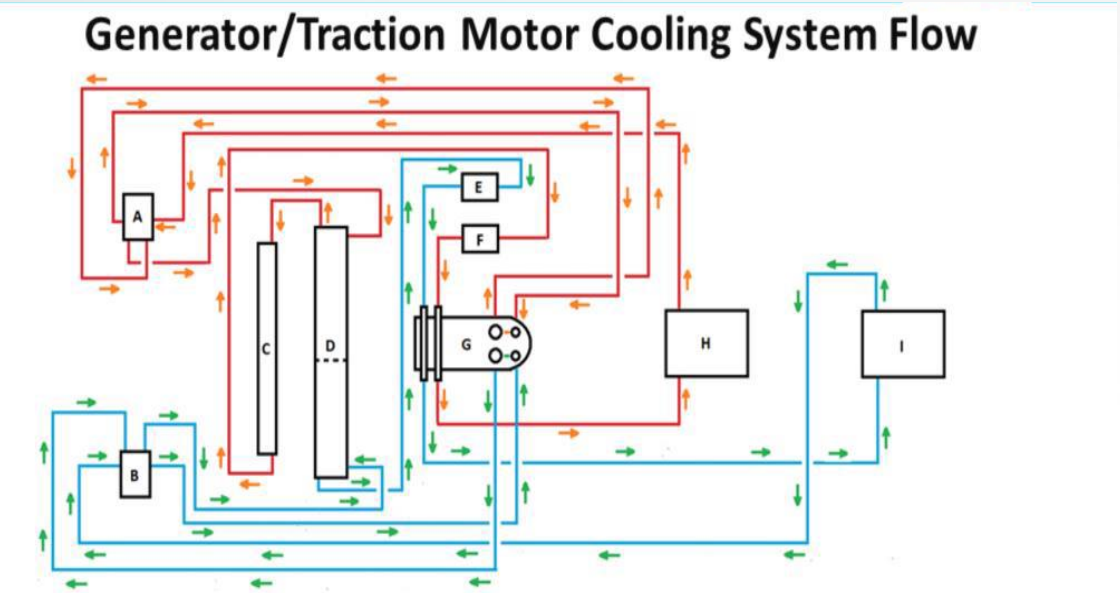
The reservoir is located on the drivers side of the vehicle near the GM windshield washer fluid fill, and should be checked weekly. To check the level of the fluid the pump must be running and can be activated with the VDSS scan tool. The fluid level should read about 2 inches below the top of the reservoir.

Warning: Steam and scalding liquids from a hot cooling system can blow out and cause serious burns. Never turn the cap when the cooling system is hot. Wait for the cooling system and reservoir tank pressure cap to cool.

Note: Blue lines to the HV battery are for a vehicle option that may not be present on every vehicle.

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GENERATOR AND TRACTION MOTOR COOLING SYSTEM



The Generator and Traction Motor Cooling Systems use DEXRON VI Automatic transmission fluid and are controlled by separate pumps but share the same reservoir. The system is comprised of, 2 coolant pumps Traction (located on the driver side) and Generator (located on the passenger side), a reservoir, 2 inline filters, 2 radiators, a pressure sensor for each loop, and cooling fans.

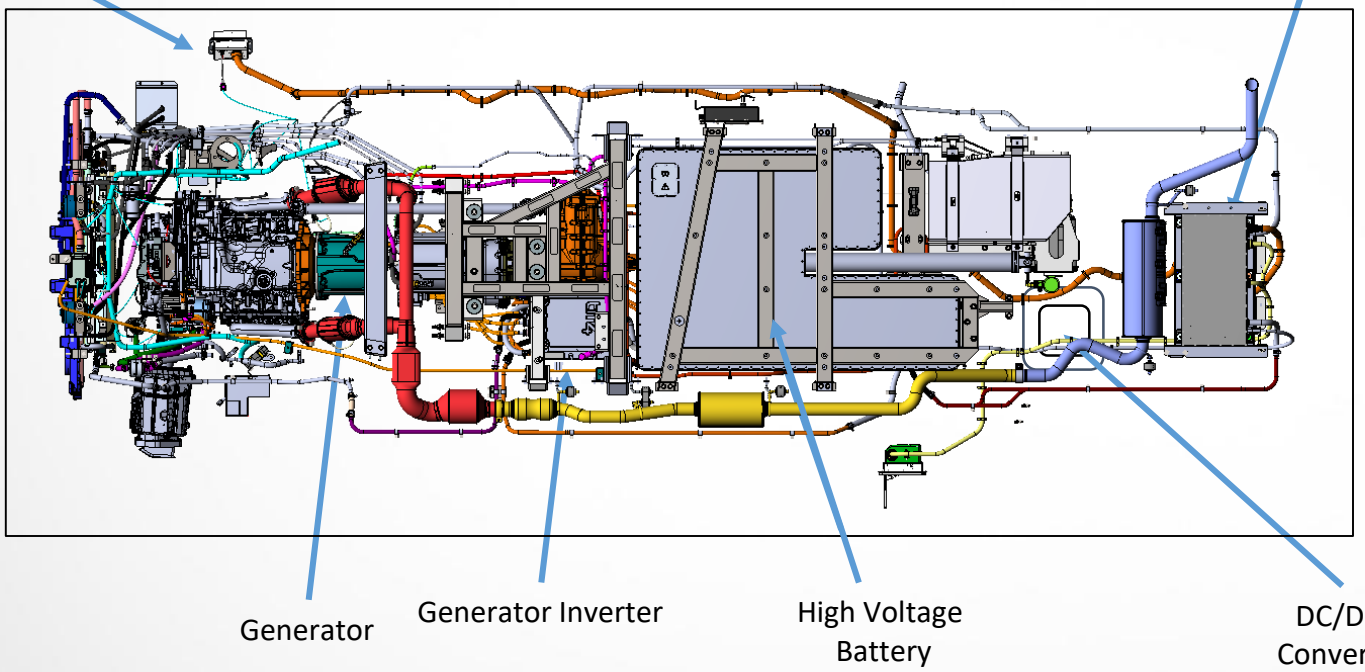
The reservoir is located in the front drivers side corner behind the headlight in the engine bay. Fluid should be checked weekly with the vehicle off when fluid is cold. Fluid should read at the top mark on the dipstick. Total system capacity is approximately 10.5 U.S. Quarts.

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CHARGING SYSTEM

The charging system is made up of the following components:

Charging System



- Electric Vehicle Supply Equipment Port or EVSE
- Bi-directional charger
- High Voltage Battery
- Generator
- Generator Inverter
- DC to DC Converter

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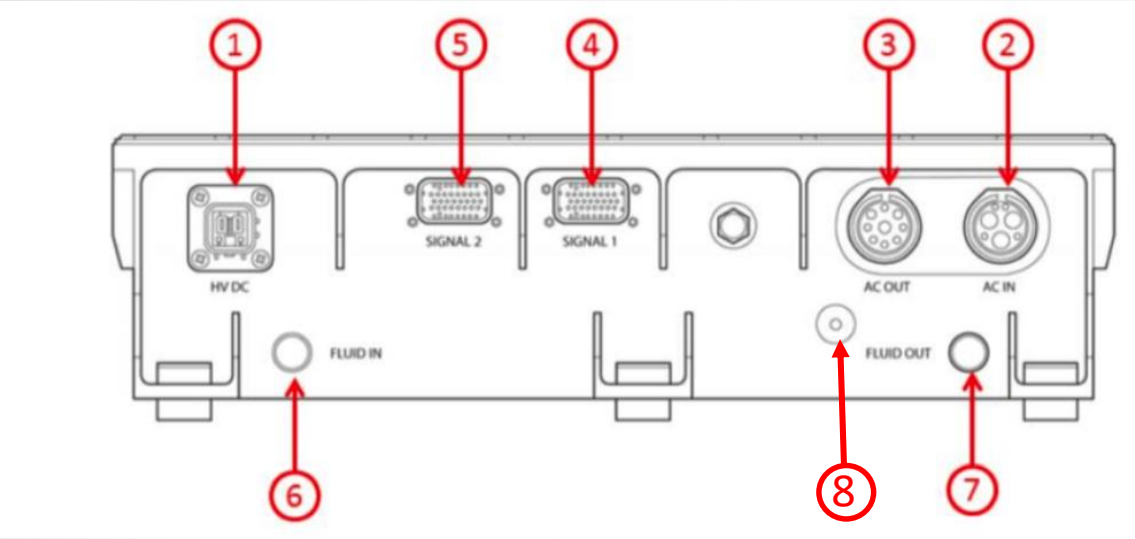
EVSE: ELECTRICAL VEHICLE SUPPLY EQUIPMENT PORT

- The Electric Vehicle Supply Equipment (EVSE) port is part of the Charging System and is located on the left front fender.
- Level 1 (120 volt) or level 2 (240 volt) EVSE charger can be used. A/C voltage is sent to the Bi-directional charger.
- The power electronics pump is located near the center of the vehicle above the front brush guard. It will run during the external charging process and can usually be heard.
- Additional clicking sounds are caused by the electrical devices coming online during charging and are normal to hear.
- The EVSE port indicator lamp will illuminate under these conditions:
  - Flashing green light indicates A/C power is flowing to charge the battery.
  - Solid green light indicates that the charge is complete.
  - Solid red light indicates a charging system fault and the vehicle is not currently charging.

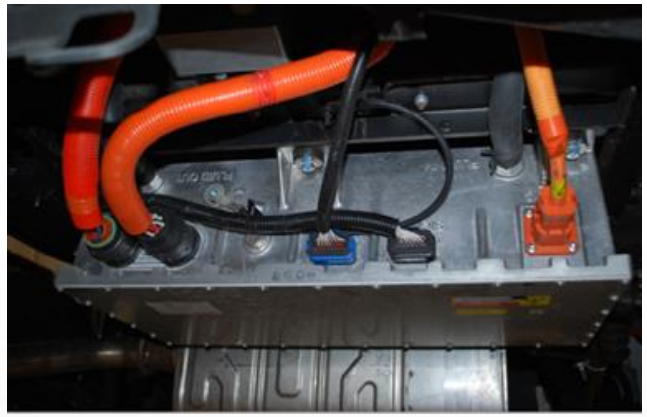


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BI-DIRECTIONAL CHARGER



1.HV connection goes to the HV battery (DC high voltage)
2. HV connection goes to the EVSE charging connection (AC voltage)
3. Export power panel connection (AC voltage)
4. Export power panel control connection
5. Bi-directional charger control connector
6. Power electronics coolant inlet hose
7. Power electronics coolant outlet hose
8. Ground



- The Bi-directional charger is located under the rear of the vehicle in front of the muffler.
- It performs two functions:
  - Converts either 120 or 240 volt A/C power from the EVSE into D/C current to charge the high voltage battery.
  - Converts D/C voltage from the High voltage battery into 240 volt A/C voltage to supply the export power panel.
- The Bi-directional charger operates from 240 to 400 volts D/C and 120 to 240 volts A/C . The bi-directional charger powers up to 14.4 kilowatts in either direction.
- The charger is liquid cooled with a 50-50 mixture of DEX-COOL and deionized water.

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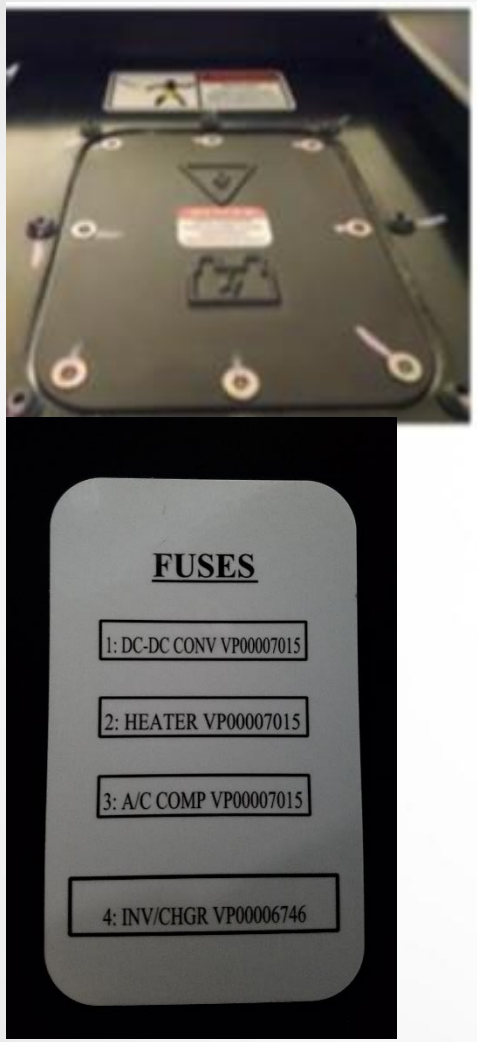
HIGH VOLTAGE BATTERY PACK

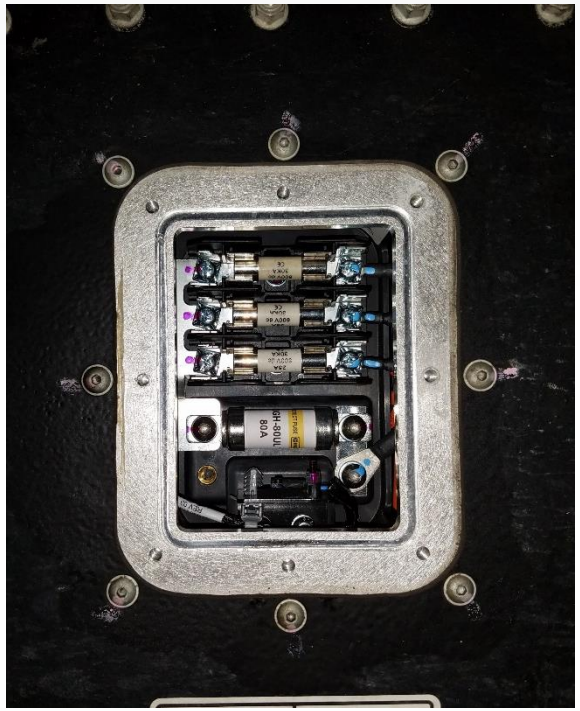


- The high voltage battery is located between the H-frame and the rear differential.
- The high voltage battery weighs approximately 800 pounds.
- Power is provided by a lithium ion high voltage battery pack with up to 400 volts available.
- The high voltage battery supplies high voltage to:
  - DC to DC converter
  - Traction Inverter
  - Generator Inverter
  - Bi-directional charger
  - AC Compressor
  - Cabin Heater
- The Manual Service Disconnect (or MSD) disconnects the High Voltage Intercept Loop (or HVIL) so the pack will not receive a 12 volt signal and mechanically disconnects the high voltage system.
- There are no serviceable components inside the battery pack; only replaceable system fuses within the battery pack fuse center.

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BATTERY PACK FUSE CENTER





The fuse center for the high voltage system is located on the underside of the HV battery.

This contains the following fuses:

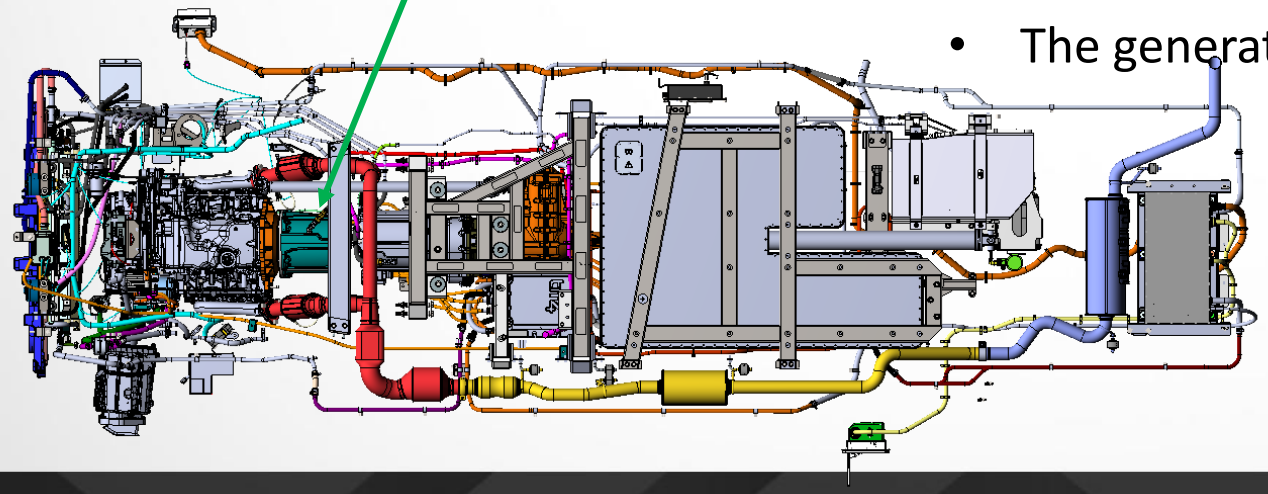
- DC/DC converter fuse (25 amp)
- Cabin heater fuse (25 amp)
- A/C compressor fuse (25 amp)
- Bi-directional charger fuse (80 amp)

WARNING: High Voltage precautions should be taken before servicing the fuses. Refer to High Voltage safety training.

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GENERATOR



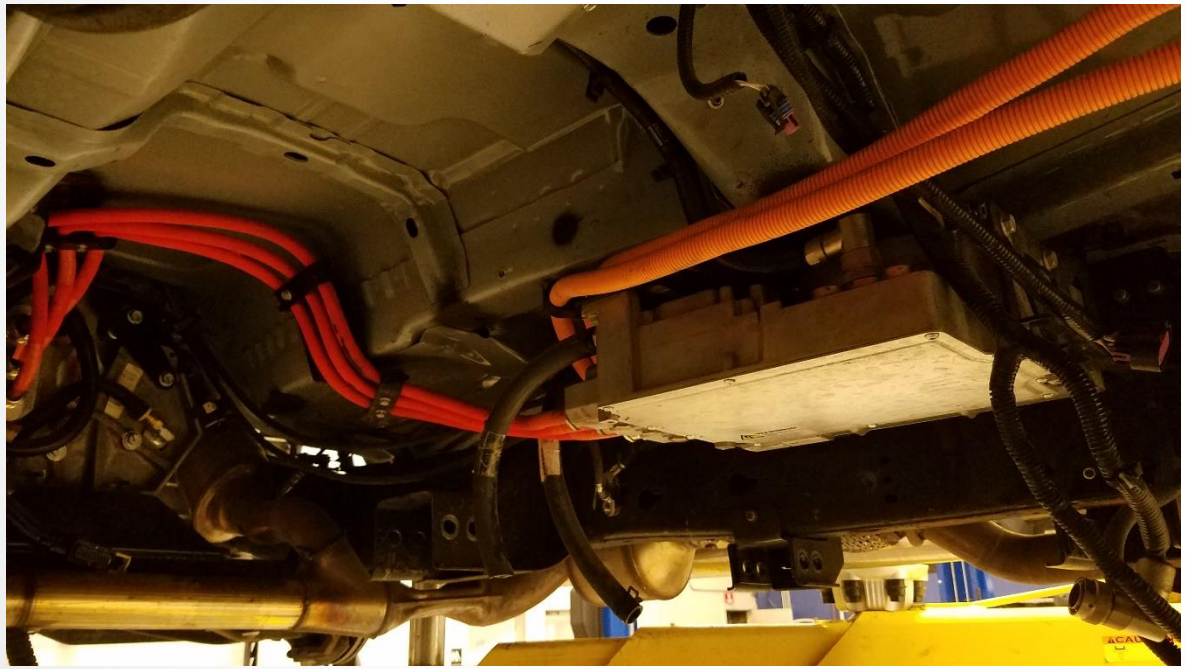


- The generator is part of the charging system and is mounted to the rear of the Internal Combustion Engine (ICE)
- The generator is controlled by the generator inverter.
- The generator uses a dry sump lubrication system with DEXRON VI Automatic Transmission Fluid.
- The generator is engine driven, and supplies current to the high voltage battery to maintain at least a 20% state of charge.
- The generator is used to start the Internal Combustion Engine.
- The generator is used to produce A/C power.

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GENERATOR INVERTER

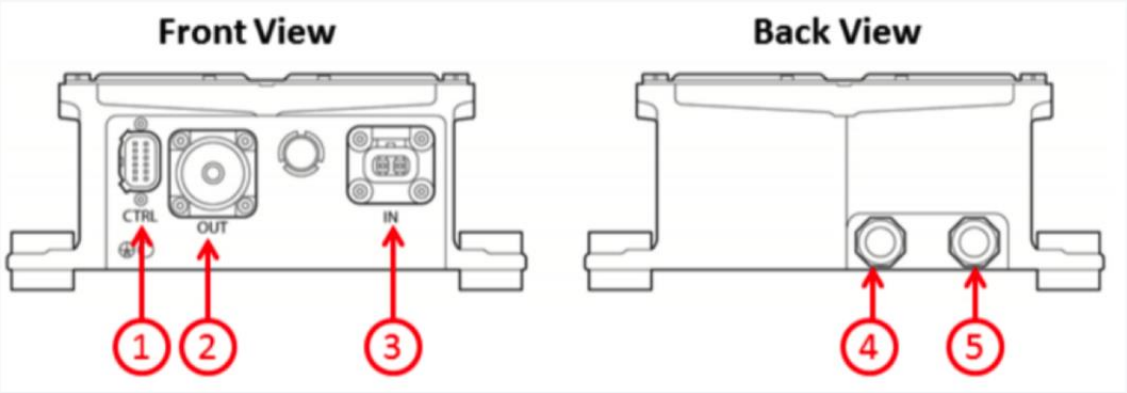
View of inverter with H-frame removed



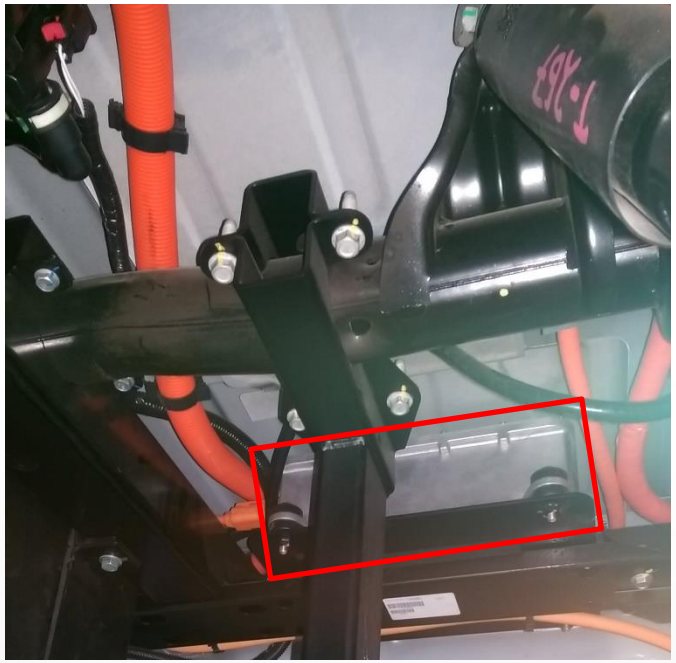
- The generator inverter is mounted to the vehicle, next to the right inner frame rail, above the traction motor inverter.
- The H-frame must be removed to service this component.
- The generator inverter converts A/C voltage to D/C voltage to charge the High Voltage battery.
- The generator inverter allows the output from the generator to be controlled based on demand.
- The generator output supplies power to maintain the high voltage battery state of charge.
- Automatically runs the Internal Combustion Engine (ICE) when high voltage battery reaches a 20% state of charge to maintain charge.
- The generator inverter is liquid cooled by a mixture of 50% DEX-COOL and deionized water.

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DC TO DC CONVERTER



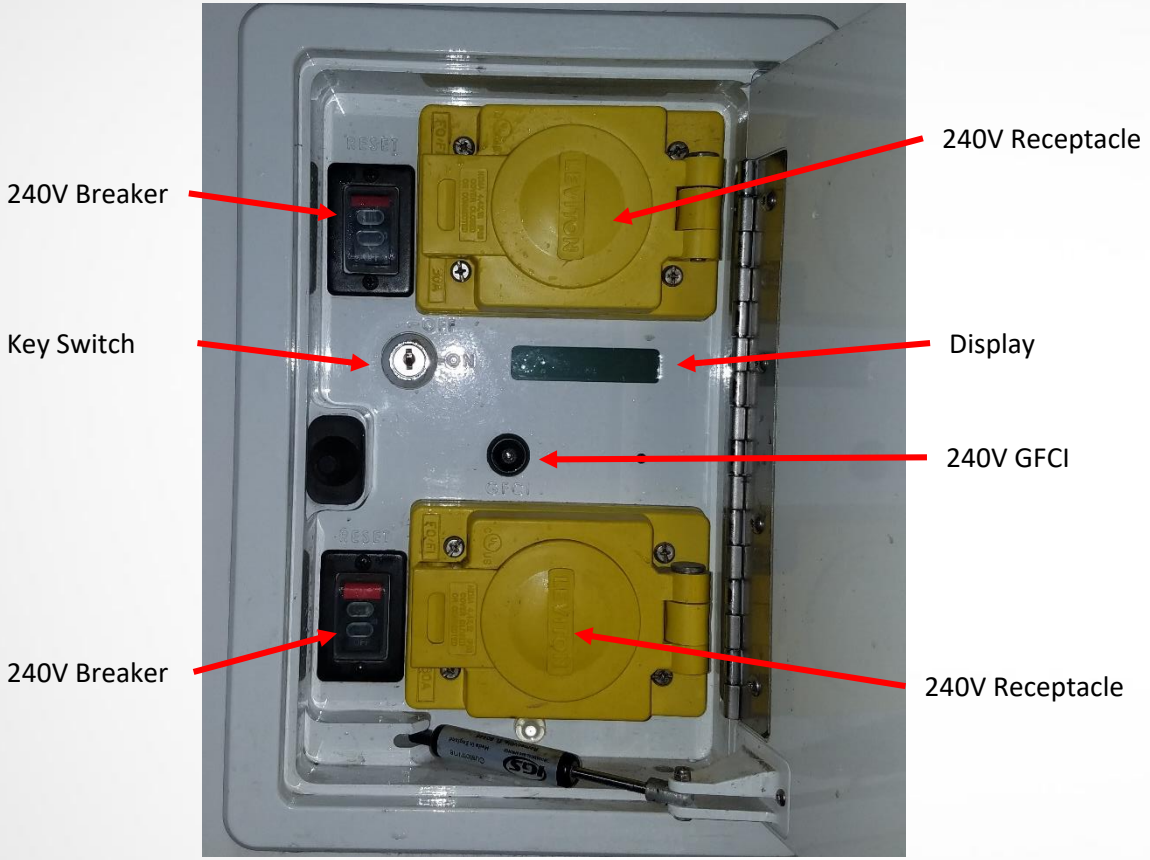
1. DC/DC converter control connector
2. Low voltage output (DC voltage) to LV battery
3. High voltage input (DC voltage) from HV battery
4. Power electronics coolant outlet hose
5. Power electronics coolant inlet hose



- The DC to DC converter is located above a cross member just above the long tail of the high voltage battery (shown in red).
- It reduces the voltage of the high voltage battery to system voltage which will:
  - Recharge the vehicle's 12 volt battery.
  - Provide voltage to maintain the vehicle 12 volt systems while operating replacing the factory alternator.
- The input voltage ranges from 240 to 400 volts, with a nominal 350 volts.
- The output voltage ranges from 12 to 15 volts with a nominal voltage of 14.4 volts.
- The DC to DC converter is liquid-cooled uses a mixture of 50% DEX-COOL and deionized water.

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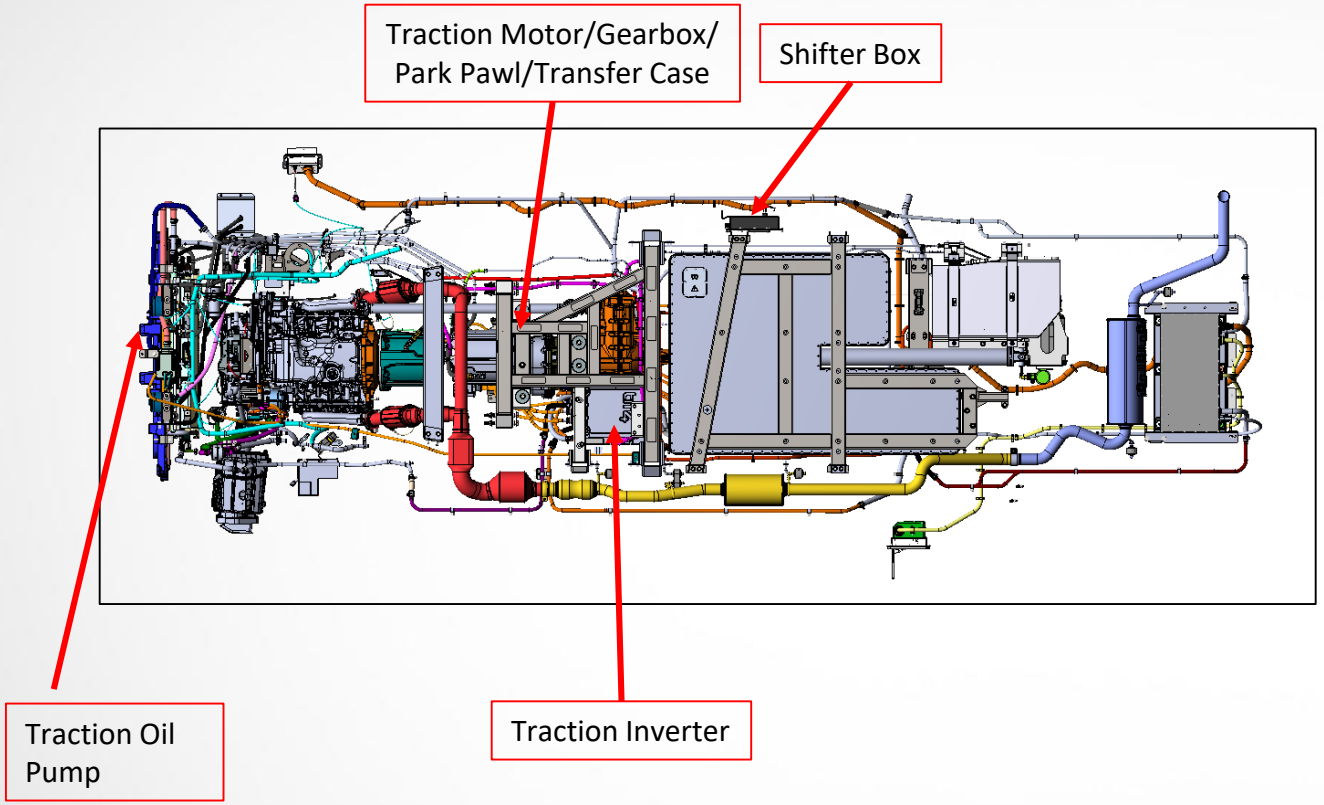
EXPORT PANEL



- The export power panel is located on the passenger side of the vehicle bed between the cab and rear tire.
- It contains two 30 amp, 240 volt A/C outlets: One 3 prong and one 4 prong
- An adapter may be used to provide two 120v outlets.
- There is 14.4 kilowatts of total available power between both connectors.
- To use export, insure the ignition key is in the off position and the charger is unplugged. Turn the export key switch to the “ON” position and switch the breaker to the “ON” position, voltage will now be available at the 240v receptacle.
- Export power may be used until the high voltage battery reaches a 20% state of charge. At that point an alarm will sound and the outlet will shut down.
- For continued export use, insert the vehicle key into the ignition and turn to the run position. The engine will start and drive the generator for continued power.

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DRIVE SYSTEM

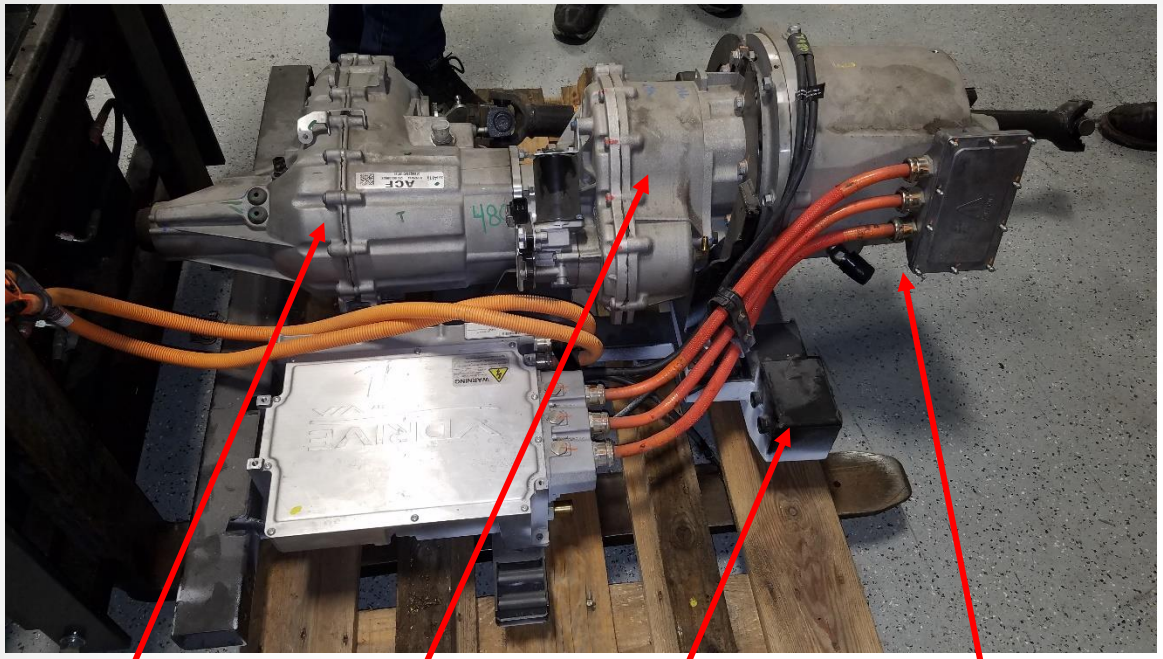


The traction (or drive) system is made up of the following components:

- Traction motor
- Traction inverter
- Shifter Box (PRNDL)
- Gear box and Park pawl motor

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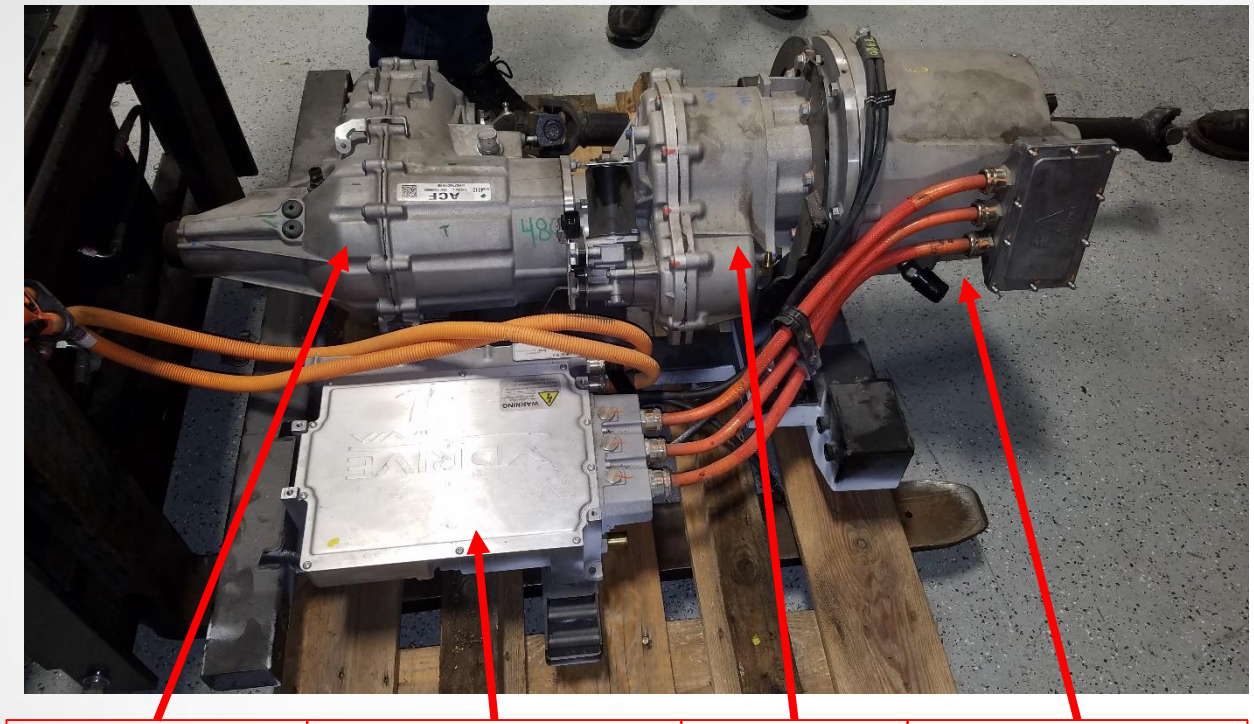
TRACTION MOTOR



- The traction motor is located on the H-frame and is bolted to the gearbox.
- The Transfer case is mounted to the back of the gearbox.
- The H-frame must be removed to service the traction motor.
- The traction motor is controlled by the traction inverter.
- It uses a dry sump lubrication system with DEXRON VI automatic transmission fluid.
- The traction motor will either use or produce A/C current.
- It uses power during propulsion and produces power during regenerative braking.

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TRACTION INVERTER



- The traction inverter is located on the passenger side of the H-frame below the generator inverter.
- It converts DC power from the high voltage battery to regulated 3-phase A/C current.
- It controls high voltage to the traction motor during acceleration.
- Regenerative braking enables the traction motor to operate as a generator when coasting or braking.
- The traction inverter is liquid-cooled uses a mixture of 50% DEX-COOL and deionized water.

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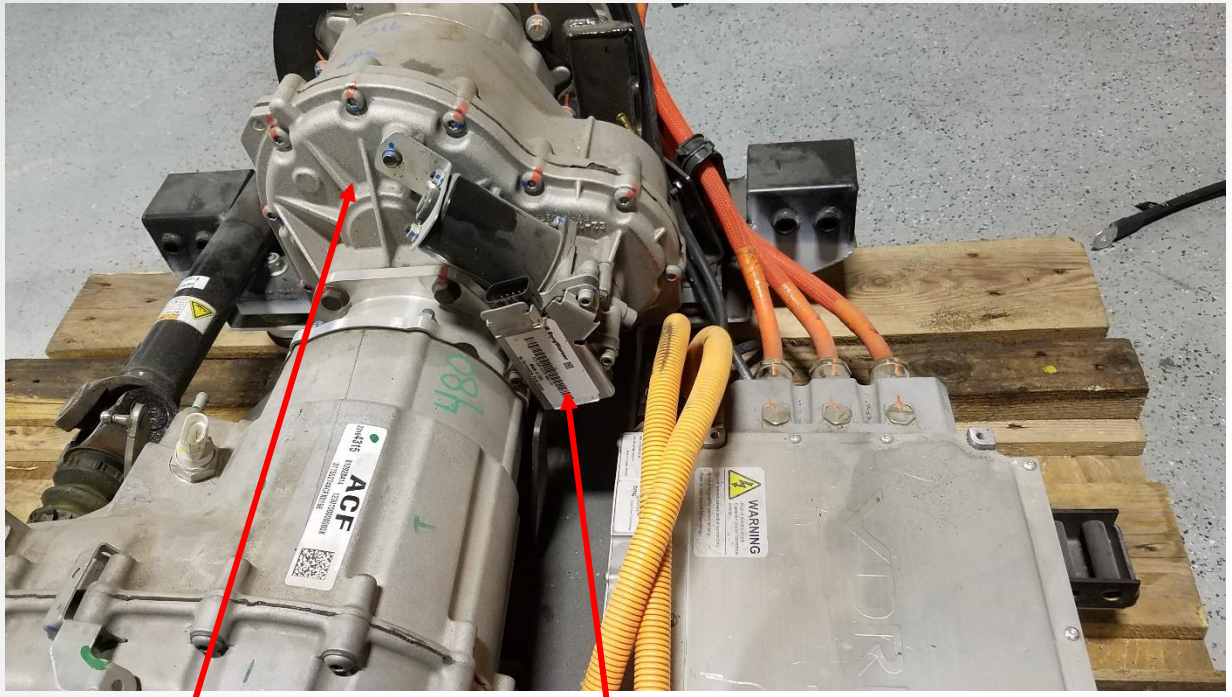
SHIFT BOX (PRNDL)



- The Shift Box (PRNDL) is located near the center of the vehicle on the drivers side outer frame rail.
- Receives manual input from the gear selector and converts it to an electric signal.

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GEARBOX AND PARK PAWL MOTOR

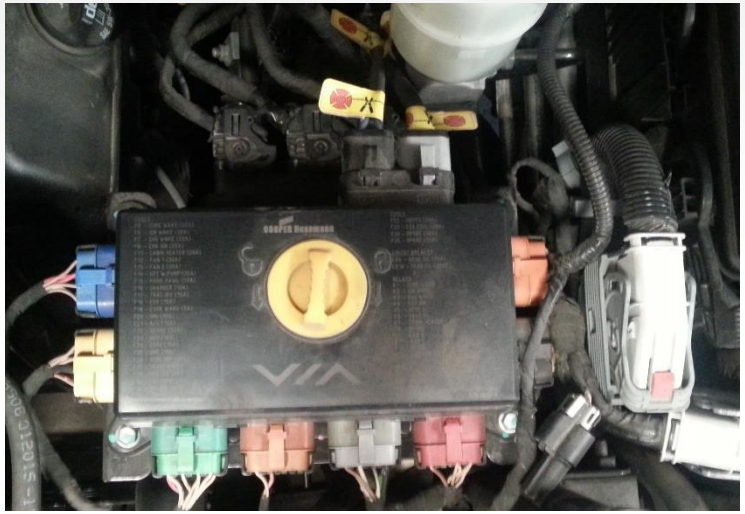


- The gear box is located on the H-frame mated to the traction motor.
- The H-frame must be removed to service the gear box.
- The gear box is a single speed, 2.68 to 1 gear reduction, used in conjunction with the traction motor to deliver power to the transfer case and differential.
- The gear box uses a 12 volt E Park module (or park pawl motor) to engage and disengage the park pawl inside the gear box.
- It uses 1.4 liters of DEXRON VI automatic transmission fluid.
- Park pawl motor locks the vehicle into park.
- The park pawl motor can be serviced without removing the H-frame.

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LOW VOLTAGE DISTRIBUTION BOX AND HYBRID CONTROL UNIT(HCU)

Bussmann Low Voltage Distribution Box



Hybrid Control Unit

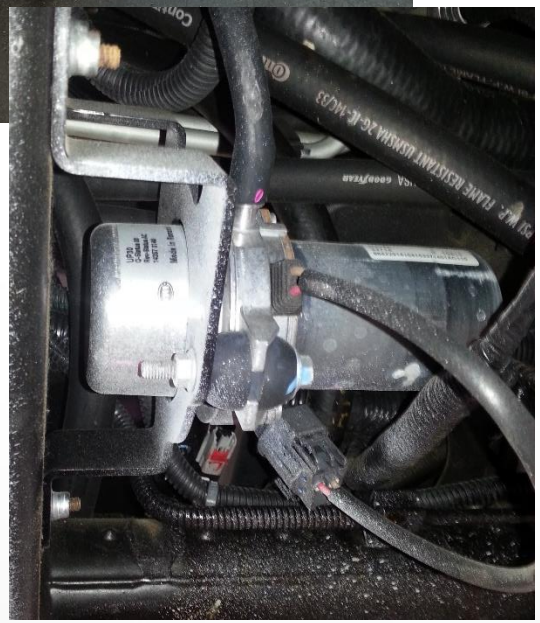


- The Low Voltage Distribution Box is located under the hood on the drivers side.
- The fuse locations are inscribed on the fuse center cover.
- The Hybrid Control Unit or HCU is located directly under the fuse box.
- The HCU manages input and signals from both GM and VIA systems for total vehicle operation.

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HVAC SYSTEM





- A High Voltage air conditioning compressor has been added to replace the GM factory belt driven compressor.
- This system uses the remaining GM factory A/C system.
- The quantity of oil will remain the same for the system
- The electric pump uses POE compressor oil and not the traditional PAG oil

- A vacuum pump has been added as the factory belt driven pump cannot operate while the vehicle is in HV mode.
- The pump is located on the inside of the frame rail near the front cross support.
- The electric vacuum pump supplies vacuum to the brake booster.

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HVAC SYSTEM





- A high voltage cabin heater has been added. Located on the cross member under the engine. It heats the coolant for cabin when the internal combustion engine (ICE) is not operating.
- A coolant temperature sensor is located near the outlet of the heater and is used to monitor the coolant output temperature of the heater.
- Amperage to the heater is controlled to vary the temperature output.
- A variable speed coolant pump is also added to the system
- The coolant pump pumps the heated coolant through a thermostat and into the GM coolant loop to the heater core when the ICE is not operating.

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MAINTENANCE INTERVALS

Service	25,000 mi / 40,000 km	50,000 mi / 80,000 km	75,000 mi / 120,000 km	100,000 mi / 160,000 km	125,000 mi / 200,000 km	150,000 mi / 240,000 km
Generator / Traction Motor Coolant		✓		✓		✓
Generator / Traction Motor Coolant Filters	✓	✓	✓	✓	✓	✓
Power Electronics Coolant						✓
Visually inspect the accessory drive belt [1]						✓
Internal Combustion Engine (ICE) Coolant [2]						✓
Internal Combustion Engine Oil	Changed when the "change engine oil soon" message is displayed or once a year, whichever comes first.					

[1] Or every 10 years, whichever comes first. Inspect for fraying, excessive cracking or damage and replace, if needed.
[2] Or every 5 years, whichever comes first

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